Introduction to Quantitative Equational Logic

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Abstract

We provide an overview of quantitative equational logic, an adjustment of the familiar equational logic system that now allows us to reason quantitatively. We first discuss model theory in first-order logic and a generalization to arbitrary logic systems called abstract model theory. We then introduce equational logic and its completeness theorem, and then we introduce quantitative equational logic by explaining the relevant adjustments made to equational logic and their consequences. Finally, we provide some motivation for the usefulness of this new system by explaining its striking connection to probability.

1 Introduction

Model theory is a common way to assign semantics to a syntactic logic system. It's most commonly studied under first-order logic due to the richness of results but we give a brief extension to abstract model theory.

Equational logic is a syntactic system that studies formal equations and their relations and helps with reasoning about programs. **Algebras** are used for their semantics and how to reason about the satisfaction of judgements. Equational logic is effective in modelling traditional algebraic structures like groups, rings, and vector spacesb but cannot be used to reason quantitatively.

Quantitative equational logic was introduced to fill this gap. Similar in nature to equational logic in many ways, quantitative equational logic differentiates itself by using pseudometrics to *approximate* equality rather than equivalence classes used in equational logic. Remarkably, certain quantitative equational algebras with metrics over probability spaces are equivalent to the freely generated algebras.

2 Model Theory

2.1 Classical Model Theory

Model theory is in essence a formal way one can assign interpretations or semantics to syntactic logic systems. In this section, we will focus on the model theory of first-order logic (FOL), which one may consider "classical model theory" as a reference. Much of the tools in equational theory have much overlap and utilize model theoretic ideas, such as what truth means in an algebra and completeness theorems. We provide some background on model theory in the following and give connections about its importance later.

First, we give the basics of the syntax of our logic system. We will use the logical symbols in first-order

logic (FOL), such as $\wedge$, $\neg$, quantifiers $\forall$ and equality $=$, along with variables, functions of finite arity and relations of arity typically denoted by κ as in typical in FOL. These choices that define our logical syntax is simply due to the choice of FOL and could be arbitrary as we will see later in 2.2. Our main objects of concern for terms, formulas, and the models we can build out of them. The following notions are mainly due to the book [CC 90] and notes [WD15].

Definition 2.1. A **term** is one of

1. a variable
2. a constant symbol
3. an element equal to f applied to κ terms for function symbol f with arity κ
4. a finite application of the above

Definition 2.2. A **formula** is one of

1. $t_1 = t_2$ for terms t_1, t_2
2. R applied to κ terms for relation R with arity κ
3. $\neg\varphi$ for formula φ
4. $\varphi \wedge \psi$ for formula φ, ψ
5. $\forall x \varphi$ for formula φ, variable x
6. a finite application of the above

Terms are in a sense the very basic building blocks, and formulas form all the ways we can apply logic, functions, and relations to them. With these two ideas, it is natural to consider the entire space that they generate, which gives rise to languages.

Definition 2.3. A **language** $\mathcal{L}$ consists of the formulas given by a set of constants, variables, functions, and relation symbols.

One can imagine that the constants, variables, functions, and relation symbols for a "basis" of some sort for our language. This is very similar to the formal languages and their alphabets. With these notions, we can define a model for a language.

Definition 2.4. A **model** (or **structure**) $\mathfrak{A}$ for a language $\mathcal{L}$ is a tuple $\langle \mathbf{A}, \mathcal{I} \rangle$ with $\mathbf{A}$ a nonempty set called the **universe** and $\mathcal{I}$ an **interpretation function** with domain $\mathcal{L}$ where

1. if $c \in \mathcal{L}$ is a constant symbol, $\mathcal{I}(c) \in \mathbf{A}$
2. if $f \in \mathcal{L}$ is a function symbol with arity κ, $\mathcal{I}(f)$ is a function symbol with arity κ on $\mathbf{A}$
3. if $R \in \mathcal{L}$ is a relation symbol with arity κ, $\mathcal{I}(R)$ is a relation symbol with arity κ on $\mathbf{A}$

It is worth noting that model is both the universe and the interpretation function. One set may be used as the universe for different models of the same language and same goes for the interpretation function. With these definitions, we can describe what truth means for a formula in a model. Essentially, we want to check if all our notions in FOL hold for a formula (all of its possible forms under the recursive definition) in its interpretation under $\mathcal{I}$ in $\mathbf{A}$.

Definition 2.5 (Satisfiability)**.** Let $\mathfrak{A} = \langle \mathbf{A}, \mathcal{I} \rangle$ be a model, $\varphi(x_1, x_2, \cdots, x_n)$ be a formula with n free variables in language $\mathcal{L}$, and $\overline{a} = (a_1, a_1, \cdots, a_n) \in \mathbf{A}^n$. Then, $\varphi(\overline{a})$ **is true in** $\mathfrak{A}$, or we write $\mathfrak{A} \vDash \varphi(\overline{a})$ if

1. $\mathfrak{A} \vDash \varphi_1(\overline{a}) = \varphi_2(\overline{a})$ if $\mathcal{I}(\varphi_1)(\overline{a}) = \mathcal{I}(\varphi_2)(\overline{a})$
2. $\mathfrak{A} \vDash R(\varphi_1(\overline{a}), \cdots, \varphi_n(\overline{a})$ if $(\mathcal{I}(\varphi_1)(\overline{a}), \cdots, (\mathcal{I}(\varphi_n)(\overline{a}) \in \mathbf{A}^n$
3. $\mathfrak{A} \vDash \phi_1(\overline{a}) \wedge \phi_2(\overline{a})$ if $\mathfrak{A} \vDash \phi_1(\overline{a})$ and $\mathfrak{A} \vDash \phi_2(\overline{a})$
4. $\mathfrak{A} \vDash \neg\phi(\overline{a})$ if not $\mathfrak{A} \vDash \phi(\overline{a})$
5. $\mathfrak{A} \vDash \forall x \phi(a_1, \cdots, a_{n-1}, x)$ if for all $a_n \in \mathbf{A}$, $\mathfrak{A} \vDash \phi(\overline{a})$

We have established satisfaction for all possible recursive definitions for formulas. We can further define many useful model-theoretic notions that are analogous to ones that will follow such as theories, judgements, and completeness, but for the sake of this paper, the previous definitions are enough to motivate the relationship between Ω-algebras and models.

2.2 Abstract Model Theory

As alluded to previously, the choice of using FOL was somewhat arbitrary, even though first-order model theory is consider to be "classical" and rich with results. We could have chosen some other logic system, which is what we will define briefly here. Later on, we will make a connection with equational logic to see an application here. If we look at the definition of satisfiability and abstract away all the details, it effectively establishes a relation between models $\mathfrak{A}$ and sentences φ. With this in mind, we can define an abstraction.

Definition 2.6. An **abstract logic** L [GV05] is a 3-tuple $(M, S, \vDash)$ where $\vDash \subseteq M \times S$. We call M the **models**, S the **sentences**, and $\vDash$ the **satisfiability relation** of L.

Now, we can say a sentence $s \in S$ has model $m \in M$ if $m \vDash s$. With this notion, we can extend satisfiability to most other logic systems. Typically, a model will still consist of a universe and some interpretation function, and we will still construct the relation $\vDash$ by the whether the interpretation of a formula still holds in the universe.

3 Equational Logic

3.1 Definitions

Equational logic is a syntactic system that studies formal equations and their relations. This logic uses a syntax made up of three main components:

Function symbol: f_i,

Arity: κ_i,

Signature: $\Omega = \{f_i : \kappa_i | i \in I\}$. [Bac+20]

We say $\mathbf{T}X$ is a collection of **terms** (same as in the model theory prelude) constructed by

$$t ::== x | f(t_i)_{i \in \kappa}, \quad \text{for } x \in X \text{ and } f : \kappa \in \Omega.$$

In equational logic, the only predicate symbol is equality. There are no quantifiers. We can write an **equation** of the form

$$s = t$$

for $s, t \in \mathbf{T}X$. The set of equations over $\mathbf{T}X$ is denoted $\mathcal{E}(\mathbf{T}X)$. [Bac+20]

The notion of a **judgement** (same notion as sentences in model theory) is the relation between a *hypothesis* $\Gamma \subseteq \mathcal{E}(\mathbf{T}X)$ and a *conclusion* $\varphi \in \mathcal{E}(\mathbf{T}X)$, denoted

$$\Gamma \vdash \varphi$$

if we can conclude φ under the assumption Γ. We likewise define a **theory** to be the set of equations closed under deduction (the same as a theory in model theory, a consistent set of sentences).

3.2 Equational Theory

An **equational theory** [Bac+20] of type Ω over X is a set of judgements $\mathcal{J}$ on $\mathbf{T}X$ satisfying, for $s, t, u \in \mathbf{T}X$ and $\Gamma, \Omega \subseteq \mathcal{E}(\mathbf{T}X)$,

(**Refl**)	$\varnothing \vdash t = t \in \mathcal{J}$,
(**Symm**)	$\{s = t\} \vdash t = s \in \mathcal{J}$,
(**Trans**)	$\{s = u, u = t\} \vdash s = t \in \mathcal{J}$,
(**Cong**)	$\{s_i = t_i | i \in \kappa\} \vdash f(s_i)_{i \in \kappa} = f(t_i)_{i \in \kappa} \in \mathcal{J}$ for any $f : k \in \Omega$,
(**Subst**)	if $\Gamma \vdash s = t \in \mathcal{J}$, then $\sigma(\Gamma) \vdash \sigma(s) = \sigma(t) \in \mathcal{J}$, for $\sigma \in \Sigma(X)$,
(**Cut**)	if $\Omega \vdash \Gamma \in \mathcal{J}$ and $\Omega \vdash s = t \in \mathcal{J}$, then $\Gamma \vdash s = t \in \mathcal{J}$,
(**Assum**)	if $s = t \in \Gamma$, then $\Gamma \vdash s = t \in \mathcal{J}$.

3.3 Algebras

An **algebra** is a set of variables X with a signature of operations Ω. Defining specific operations in Ω allows us to characterize algebraic structures. We can call an **algebraic structure** $\mathcal{A} = (A, \Omega_{\mathcal{A}})$ an **Ω-algebra**, where

- A: a set of elements in the algebra, called the **carrier**, or **universe**
- $\Omega_{\mathcal{A}}$: a collection of **interpretations**, where an interpretation of a function symbol $f : \kappa \in \Omega$ is a function

$$f_{\mathcal{A}} : \underbrace{A \times \cdots \times A}_{\kappa \text{ times}} \to A$$

Notice that an algebraic structure is precisely a model for the syntactic equational logic equipped with both a universe and an interpretation function. Our constants for equational logic have been incorporated as functions with arity 0. Since we don't have relations in our logic system, our interpretation map's domain is over all functions. Our universe in this case is the same algebra A that we use for inputs to our functions since underneath the hood, we are representing homomorphisms! We can now define satisfaction in a similar manner to in 2.2.

$\mathcal{A} \vDash (\Gamma \vdash s = t)$, i.e., $\mathcal{A}$ **satisfies** the judgement $\Gamma \vdash s = t$, if for every assignment function mapping variables to their algebraic representation

$$\iota : X \to A,$$

we have

$$(\forall (s' = t') \in \Gamma,\ \ \iota(s') = \iota(t')) \implies \iota(s) = \iota(t).$$

Similar to before, satisfiability $\vDash$ establishes a relation between models $\mathcal{M}$ and judgements $\Gamma \vdash s = t$. Thus, we see this as an instantiation of our abstract logic. Underneath the hood, if we unwrap the judgment into equations, one can see that there is still the same idea of seeing whether the interpretations hold in the universe in equational logic.

3.4 Examples

Here we give some familiar examples of algebraic structures that one can construct, such as groups and barycentric algebras. The following set of variables and signature Ω capture a **group**.

$$\Omega = \{e : 0, ^{-1} : 1, \cdot : 2\}$$

$$\begin{aligned}
\textbf{(Id)} \quad & \varnothing \vdash e \cdot x = x \\
\textbf{(Id)} \quad & \varnothing \vdash x \cdot e = x \\
\textbf{(Inv)} \quad & \varnothing \vdash x \cdot x^{-1} = 1 \\
\textbf{(Assoc)} \quad & \varnothing \vdash (x \cdot y) \cdot z = x \cdot (y \cdot z)
\end{aligned}$$

To get a specific instance of a group, such as $\mathbb{Z}_4$, we need to specify a carrier, which in this case could be $\{0, 1, 2, 3\}$, and how $\cdot$ acts precisely. The following captures a **baycentric algebra**, variations of which will come in handy later.

$$\Omega = \{+_e : 2 \mid e \in [0, 1]\}$$

$$\begin{aligned}
\textbf{(B1)} \quad & \varnothing \vdash x +_1 y = x \\
\textbf{(B2)} \quad & \varnothing \vdash x +_e x = x \\
\textbf{(SC)} \quad & \varnothing \vdash x +_e y = y +_{1-e} x \\
\textbf{(SA)} \quad & \varnothing \vdash (x +_{e_1} y) +_{e_2} z = +_{e_1 e_2}(y +_{\frac{e_2 - e_1 e_2}{1 - e_1 e_2}} z) \ \text{ for } \ e_1, e_2 \in (0, 1)
\end{aligned}$$

It's worth that noting that most familiar algebraic objects, such as groups, rings, semigroups, monoids, vector spaces, can all be expressed with equational logic. However, we are unable to capture inequality, which means fields in a traditional sense can not be expressed with equational logic (since we can not capture if $x \neq 0$, then x has an inverse). However, there is a workaround referred to as a **meadow**

[BHT09], which we are showing here merely for interest. We start with the signature for a commutative ring along with its properties.

$$\Omega = \{1 : 0, 0 : 0, - : 1, + : 2, \cdot : 2\}$$

Then, instead of adding mulitplicative inverses as one normally would to get a field, we add total inverse, which is an operation $^{-1} : 1$ added to Ω and new properties

$$\varnothing \vdash (x^{-1})^{-1} = x$$
$$\varnothing \vdash x \cdot (x^{-1} \cdot x) = x$$

to obtain a meadow, where inverses are the same as in field except one can derive $0^{-1} = 0$.

3.5 Completeness

Birkhoff's completeness theorem tells us that equational logic with a standard proof system is complete, i.e., all true statements are provable.

Theorem 3.1 (Completeness, Birkhoff)**.**

$$\mathcal{U} \vDash (\Gamma \vdash \varphi) \textit{ if and only if } (\Gamma \vdash \varphi) \in \mathcal{U}$$

where $\mathcal{U}$ is a theory, and $\mathcal{U} \vDash (\Gamma \vdash \varphi)$ means $\mathcal{A} \vDash (\Gamma \vdash \varphi)$ for any algebra $\mathcal{A}$ that satisfies all the judgements in $\mathcal{U}$.

The proof of this statement is outside the scope of this paper and can be found in the chapter [Bac+20].

Considering that equational logic is a system that consists of quantifier-free symbols from first-order logic, the completeness of equational logic is remarkable in that it tells us that a valid equational logic formula φ, provable using a set of judgements (i.e., an equational theory) $\mathcal{U}$ in equational logic, is also provable in first-order logic.

4 Quantitative Equational Logic

The exact same model theoretic concepts can be extended for quantitative equational logic similar to equational logic, so we will not repeat them here. However, we hope we demonstrated the flexibility of model theory, particularly in its more abstract form.

4.1 Definitions

Quantitative equational logic studies analogous notions to equational logic, to which we extend the notion of an equation new idea known as a **quantitative equation**. We write

$$s =_\varepsilon t$$

to mean "s is within ε of t."

To formalize the notion of $=_\varepsilon$, we say the above is a quantitative equation of $s, t \in \mathbf{T}X$, with $\varepsilon \in \mathbb{Q}^+$[1]. It is *not* an equivalence relation, as transitivity immediately breaks down; for example, $\{x =_1 y, y =_1 z\} \nvdash x =_1 z$.

Quantitative equational reasoning mimics equational logic with $=_0$, representing strict equality. We can say that quantitative equations generalize equations. If we let the set of quantitative equations be $\mathcal{I}(\mathbf{T}X)$, then it is easy to see that $\mathcal{E}(\mathbf{T}X) \subseteq \mathcal{I}(\mathbf{T}X)$.

4.2 Quantitative Equational Theory

As noted previously, the transitivity of equational theory no longer holds in quantitative equational theory when $\varepsilon \neq 0$, which means a quantitative equation is *not* an equivalence relation.

We replace the notion of transitivity with a rule related to the *triangle inequality*, which is often stated as

$$|a + b| \leq |a| + |b|.$$

We translate this to a rule,

$$\textbf{(Triang)}\ \{s =_\varepsilon u, u =_{\varepsilon'} t\} \vdash s =_{\varepsilon+\varepsilon'} t \in \mathcal{J}.$$

There are other rules for quantitative equational theory analogous to those of equational theory (including **Subst**, **Cut**, and **Assum**) with appropriate adjustments, as well as others that follow logically from application of the triangle inequality.

4.3 Quantitative Algebras

Recall that an **algebra** is a set of variables X with a signature of operations Ω and that defining specific operations in Ω allows us to characterize algebraic structures.

We can call an algebraic structure $\mathcal{A} = (A, d, \Omega_\mathcal{A})$ an Ω-algebra, with

- A: a set of elements in the algebra, called the **carrier**
- d: metric for extended metric space (A, d)
- $\Omega_\mathcal{A}$: a collection of **interpretations**, where an interpretation of a function symbol $f : \kappa \in \Omega$ is a **nonexpansive function**

$$f_\mathcal{A} : \underbrace{A \times \cdots \times A}_{\kappa \text{ times}} \to A$$

Note that this is equivalent to that of equational algebras but with a metric.

[1] ε being a positive rational number is not necessarily strict; in fact, in some cases it can instead be interpreted as a positive real number, and also ∞.

4.4 Examples

Equipped with the power of quantitative equational logic, we can define barycentric algebras in the exact same manner but replacing = with $=_0$ as noted before. We will restate it here for completeness.

Example 4.1 (Barycentric Algebra (quantified))**.**

$$\Omega = \{+_e : 2 \mid e \in [0,1]\}$$

(B1) $\varnothing \vdash x +_1 y =_0 x$
(B2) $\varnothing \vdash x +_e x =_0 x$
(SC) $\varnothing \vdash x +_e y =_0 y +_{1-e} x$
(SA) $\varnothing \vdash (x +_{e_1} y) +_{e_2} z =_0 +_{e_1 e_2}(y +_{\frac{e_2 - e_1 e_2}{1-e_1 e_2}} z)$ for $e_1, e_2 \in (0,1)$

To truly harness the power of quantitative equational logic, we need some properties to capture anotion of approximate equality. One useful object we can create is a **left invariant barycentric algebra**.

Example 4.2 (Left Invariant Barycentric Algebra)**.**

$$\Omega = \{+_e : 2 \mid e \in [0,1]\}$$

(B1) $\varnothing \vdash x +_1 y =_0 x$
(B2) $\varnothing \vdash x +_e x =_0 x$
(SC) $\varnothing \vdash x +_e y =_0 y +_{1-e} x$
(SA) $\varnothing \vdash (x +_{e_1} y) +_{e_2} z =_0 +_{e_1 e_2}(y +_{\frac{e_2 - e_1 e_2}{1-e_1 e_2}} z)$ for $e_1, e_2 \in (0,1)$
(LI)* $\varnothing \vdash x +_e z =_\varepsilon y +_e z$ for $e \leq \varepsilon \in \mathbb{Q} \cap [0,1]$

The first 4 axioms stay the same and the new (**LI**) roughly captures a notion of "left invariance," where the left variable in an addition doesn't change the result by more than ε. Another useful object is the **interpolative barycentric algebra**.

Example 4.3 (Interpolative Barycentric Algebra)**.**

$$\Omega = \{+_e : 2 \mid e \in [0,1]\}$$

(B1) $\varnothing \vdash x +_1 y =_0 x$
(B2) $\varnothing \vdash x +_e x =_0 x$
(SC) $\varnothing \vdash x +_e y =_0 y +_{1-e} x$
(SA) $\varnothing \vdash (x +_{e_1} y) +_{e_2} z =_0 +_{e_1 e_2}(y +_{\frac{e_2 - e_1 e_2}{1-e_1 e_2}} z)$ for $e_1, e_2 \in (0,1)$
($\mathbf{IB_m}$) $\{x =_{\varepsilon_1} y, x' =_{\varepsilon_2} y'\} \vdash x +_e x' =_\delta y +_e y'$ for $\delta \in \mathbb{Q}^+ : (e\varepsilon_1^m + (1-e)\varepsilon_2^m)^{\frac{1}{m}} \leq \delta$

Again, the first 4 axioms are still the same as per the barycentric algebra. However, now, the new ($\mathbf{IB}_m$) axiom is less intuitive. The choice of δ looks similar to an idea of exponential interpolation but surprisingly, this one captures the Kantorovich distance.

4.5 Connection with Probability Distances

Take probability measures p, q over metric space (X, d) with a Borel σ-algebra $\mathcal{B}$. Then, we can define 2 important metrics.

Definition 4.4. The **total variation metric** between p, q is given by

$$\mathrm{TV}(p,q) = \sup_{p \in \mathcal{B}} |p(E) - q(E)|$$

The total variation metric is a fairly intuitive metric that returns the maximal difference in probability when observed on all measureable events. The next distance is a bit less intuitive but still has a nice motivating explanation. We first need to define a coupling π between p, q, which intuitively represents how much mass was moved from p to q such that they match in a certain transportation.

Definition 4.5. A **coupling** π between p, q is a new probability measure over $X \times X$ such that the marginals of $\pi(p, q)$ component wise are p, q.

Definition 4.6. The **Kantorovich distance** [2] between p, q is given by

$$W^{(m)}(p,q) = \inf_{\pi \in C(p,q)} \left[\int d(x,y)^m \, \mathrm{d}\pi(x,y) \right]^{\frac{1}{m}}$$

where $C(p, q)$ is the space of couplings.

It is worth noting that the Kantorovich distance is not an actual metric since it is not symmetric. This distance is sometimes also called the earth movers' distance because intuitively if we imagine the distributions p, q as representing amount of earth on X, then it represents the minimum amount of earth needed to be moved from p to q to match when $m = 1$. The m is a generalization, similar to L^p distances. To see their connection with the quantitative equational logic theory we've developed so far, we first define a convex combination since it naturally gives rise to barycentric algebras.

Definition 4.7. A **convex combination** of probability measures $p_1, p_2, \cdots, p_n$ is $a_1 p_1 + a_2 p_2 + \cdots + a_n p_n$ where $a_i \geq 0 \forall i \in [n]$ and $\sum_{i=1}^{n} a_i = 1$.

Start with a(n) (extended) metric space (X, d) and obtain the left invariant barycentric algebra $\Pi[X]$ on the space of finitely supported distributions over X where $+_e$ is convex combinations. Then, if we give $\Pi[X]$ the total variation metric TV, we get precisely the same as the free algebra generated from X as a left invariant barycentric algebra. Therefore, this left invariant barycentric algebra somehow captures the total variation distance as it naturally arises in the construction with no mention of distances or anything similar to total variation in the definition. This same process can be repeated for interpolative barycentric algebras to give rise to the Kantorovich distance, which is what notified the authors of [MPP16] that this construction of quantitative equational logic is in fact fruitful!

4.6 Completeness

Completeness for quantitative equational logic is analogous to the completeness theorem for equational logic.

Theorem 4.8 (Completeness)**.**

$$\mathcal{U} \vDash (\Gamma \vdash \varphi) \textit{ if and only if } (\Gamma \vdash \varphi) \in \mathcal{U}$$

[2] Not Wasserstein distance! It is oftentimes miscredited to Wasserstein as he used it once years later perhaps due to Kantorovich originally using the letter W. To know more, refer to a Prakash talk (they are good!)

where $\mathcal{U}$ is a quantitative equational theory, and $\mathcal{U} \vDash (\Gamma \vdash \varphi)$ means $\mathcal{A} \vDash (\Gamma \vdash \varphi)$ for any algebra $\mathcal{A}$ that satisfies all the judgements in $\mathcal{U}$.

5 Conclusion

Quantitative equational logic expands upon equational logic in a natural manner to approximate equality, and this new logical system is shown to have powerful applications and fascinating consequences. [MPP16] develop the theory and prove completeness and show striking connections with probability theory. Extensions to their work have been explored, many of which take categorical approach to the subject in the style of Lawvere theory for the counterpart equaional logic. Markov processes have been characterized using quantitative equations in [Bac+18] as well as probabilistic programming and differential privacy with the Fuzz programming language in [Amo+19]. We are hopeful this is a rich area with much theory and applications to be developed!

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